

Utility of Minimal Physiologically Based Pharmacokinetic Models for Assessing Fractional Distribution, Oral Absorption, and Series-Compartment Models of Hepatic Clearance^S

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ABSTRACT

Minimal physiologically based pharmacokinetic (mPBPK) models are physiologically relevant, require less information than full PBPK models, and offer flexibility in pharmacokinetics (PK). The well-stirred hepatic model (WSM) is commonly used in PBPK, whereas the more plausible dispersion model (DM) poses computational complexities. The series-compartment model (SCM) mimics the DM but is easier to operate. This work implements the SCM and mPBPK models for assessing fractional tissue distribution, oral absorption, and hepatic clearance using literature-reported blood and liver concentration-time data in rats for compounds mainly cleared by the liver. Further handled were various complexities, including nonlinear hepatic binding and metabolism, differing absorption kinetics, and sites of administration. The SCM containing one to five (n) liver subcompartments yields similar fittings and provides comparable estimates for hepatic extraction ratio (ER), prehepatic availability (F_g), and first-order absorption rate constants (k_a). However, they produce decreased intrinsic clearances (CL_{int}) and liver-to-plasma partition coefficients (K_{ph}) with increasing n as expected. Model simulations demonstrated changes in intravenous and oral PK profiles with alterations in K_{ph} and k_a and with

hepatic metabolic zonation. The permeability (PAMPA P) of the various compounds well explained the fitted fractional distribution (f_d) parameters. The SCM and mPBPK models offer advantages in distinguishing systemic, extrahepatic, and hepatic clearances. The SCM allows for incorporation of liver zonation and is useful in assessing changes in internal concentration gradients potentially masked by similar blood PK profiles. Improved assessment of intraorgan drug concentrations may offer insights into active moieties driving metabolism, biliary excretion, pharmacodynamics, and hepatic toxicity.

SIGNIFICANCE STATEMENT

The minimal physiologically based pharmacokinetic model and the series-compartment model are useful in assessing oral absorption and hepatic clearance. They add flexibility in accounting for various drug- or system-specific complexities, including fractional distribution, nonlinear binding and saturable hepatic metabolism, and hepatic zonation. These models can offer improved insights into the intraorgan concentrations that reflect physiologically active moieties often driving disposition, pharmacodynamics, and toxicity.

Introduction

The liver is a pivotal eliminating organ for numerous endogenous substances and therapeutic drugs from the body. The pharmacokinetic (PK) field has sought to assess and predict hepatic clearances (CL_h) utilizing in vitro–in vivo extrapolation (IVIVE) (Jones et al., 2022; Tess et al., 2022) and physiologically based

pharmacokinetic (PBPK) modeling approaches (Taskar et al., 2021; Lin et al., 2022), where an assumed structural liver model is required. Among the well-stirred model (WSM), the dispersion model (DM), and the parallel tube model (PTM) (Bischoff and Dedrick, 1970; Rowland et al., 1973; Pang and Rowland, 1977; Roberts and Rowland, 1986a,b; Li and Jusko, 2022), the DM is considered most compatible with the known architecture and physiology of the liver. Nevertheless, its implementation in PBPK has been hindered due to the complicated secondary partial differential equations required (Roberts and Rowland, 1986a,b; Oliver et al., 2001), whereas the WSM is most adopted given its mathematical simplicity. The series-compartment model (SCM), which treats the liver as a cascade of identical well-stirred segments connected by hepatic blood flow (Q_h), reasonably approximates the DM

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ABBREVIATIONS: AIC, Akaike Information Criterion; AUC, area under the curve; AUC_{inf} , area under the curve from time 0 to infinity; B_{max} , maximum binding capacity; C_b , blood concentration; CL_{int} , hepatic intrinsic clearance; CQ, chloroquine; CyA, cyclosporine A; DLZ, diltiazem; DM, dispersion model; D_N , dispersion number; DPH, phenytoin; DZP, diazepam; EB, ethoxybenzamide; ER , extraction ratio; f_d , fractional distribution; F_g , prehepatic bioavailability; F_{po} , oral bioavailability; FTY720, fingolimod; f_{ub} , unbound fraction in blood; f_{up} , unbound fraction in plasma; GI, gastrointestinal; HB, hexobarbital; k_a , absorption rate constant; K_d , equilibrium dissociation constant; K_m , Michaelis-Menten constant; K_{ph} , liver-to-plasma partition coefficient; mPBPK, minimal physiologically based pharmacokinetic; n , number of liver subcompartments; NIC, nicotine; P , permeability coefficient; PAMPA, parallel artificial membrane permeability assay; PBPK, physiologically based pharmacokinetic; P-gp, P-glycoprotein; PK, pharmacokinetics; PP, periportal; PPN, propranolol; PTZ, pentazocine; PV, perivenous; QD, quinidine; Q_h , hepatic blood flow; R_b , blood-to-plasma ratio; SCM, series-compartment model; TB, tolbutamide; T_{max} , time to reach C_{max} ; VEM, verapamil; V_{max} , maximum rate of metabolism; WSM, well-stirred model.

(Roberts and Rowland, 1986a; Gray and Tam, 1987; Anissimov and Roberts, 2002). Subsequently, variants of the SCM were applied to mimic the DM for characterizing the hepatic disposition of transporter substrates (Watanabe et al., 2009; Jones et al., 2012; Li et al., 2014; Morse et al., 2017) and predicting the clinical drug-drug interactions (DDIs) involving hepatic transporters/enzymes (Asaumi et al., 2019; Zhang et al., 2022). The quantitative correlation of the SCM and DM was not elucidated until our recent work (Li and Jusko, 2023), where we demonstrated how the number of liver subcompartments (n) in the SCM correlated with the dispersion number (D_N) of the DM that describes the relative spread of solute on passage through the liver. We also showed that the SCM closely resembles the DM in PBPK with easier operative features and offers many flexibilities such as intraorgan concentration gradients along the sinusoidal path of the liver and zonal heterogeneity of enzymes and transporters.

Minimal physiologically based pharmacokinetic (mPBPK) models, introduced for characterizing and predicting the pharmacokinetics (PK) of both small (Cao and Jusko, 2012) and large molecules (Cao et al., 2013), inherit and lump major physiologic attributes from the whole-body PBPK models yielding physiologically relevant PK parameters (Jeong et al., 2022a,b) and thereby provide more practical assessments than do mammillary models. Owing to their physiologic relevance; flexibilities in handling different mechanisms pertaining to the absorption, distribution, metabolism, and excretion processes; easier operative features; and less information required than full PBPK models, mPBPK models have been progressively applied in both preclinical and clinical settings. These include: target-mediated drug disposition (Cao and Jusko, 2014; Chen et al., 2016), the neonatal Fc receptor recycling mechanism (Hardiansyah and Ng, 2022) of large molecules, interspecies PK scaling of small (Cao and Jusko, 2012; Jeong and Jusko, 2021; Song and Jusko, 2021) and large molecules (Zhao et al., 2015; Chen et al., 2017), the nonlinear binding PK of naproxen (Li et al., 2017), sex differences in hepatic metabolism of methylprednisolone (Ayyar et al., 2019), drug-drug-disease interactions of rivaroxaban and verapamil in subjects with varying degrees of renal impairment (Ismail et al., 2018), anti-inflammatory effects of dexamethasone (Świerczek and Jusko, 2023), predicting lung exposure of ivermectin for COVID-19 drug repurposing (Jermain et al., 2020), brain distribution of the antibiotic metronidazole (Chauzy et al., 2022), antibody therapies (Bloomingdale et al., 2021) targeting central nervous system (CNS) disorders, the kinetics of novel treatment modalities including small interfering RNA (siRNA) (Ayyar et al., 2021) and chimeric antigen receptor (CAR) T cells (Tsai et al., 2022), and the PK of antenatal betamethasone and dexamethasone in parturient women and their fetuses (Krzyszanski et al., 2023).

Considering the advantages of mPBPK models and evidence that the SCM closely approximates the DM in PBPK, this work demonstrates applications of the SCM in mPBPK and further expands the utility of mPBPK models for: 1) assessing hepatic distribution and clearance; 2) characterizing the oral absorption kinetics and first-pass effects (prehepatic and hepatic extraction), utilizing literature-reported single intravenous (i.v.) and oral blood, plasma, and liver concentration-time data in rats for several flow-limited compounds that are primarily eliminated through the liver; and 3) relating the fractional distribution (f_d) parameter to tissue permeability. The changes in the blood and liver PK after intravenous or oral dosing due to alterations in drug- or biologic system-specific factors were also explored.

Methods

Data Sources. The observed blood or plasma and liver concentration-time data in rats for 14 compounds [i.e., tolbutamide (TB), chloroquine (CQ), cyclosporine A (CyA), ethoxybenzamide (EB), fingolimod (FTY720), nicotine (NIC), pentazocine (PTZ), hexobarbital (HB), phenytoin (DPH), quinidine (QD), propranolol (PPN), diazepam (DZP), verapamil (VEM), and diltiazem (DLZ)] were digitized from the references listed in Table 1 using GetData Graph Digitizer version 2.26 (<https://getdata-graph-digitizer.software.informer.com/>). For the selected compounds: 1) liver is the major eliminating organ and the extrahepatic clearances are known or assumed to be negligible; 2) distribution into the liver and access to the hepatic enzymes are flow limited (high permeability) with minor or negligible transporter involvements; and 3) hepatic extraction ratios (ER) range from low to high. The reported plasma concentrations (C_p) were first converted to the blood concentrations (C_b) by multiplying the blood-to-plasma ratio (R_b) (Supplemental Table 3) before fitting.

Model Structures. The minimal PBPK model with one or two tissue compartments extended with the SCM of hepatic elimination is shown in Fig. 1. The SCM has been described in detail (Li and Jusko, 2023). Briefly, the liver is divided into n well stirred subcompartments connected by hepatic blood flow (Q_h), and the tissue volume (V_{hi}), availability (F_i), extraction ratio (ER_i), and intrinsic clearance ($CL_{int,i}$) for each subcompartment i ($i = 1, 2, 3 \dots n$) are initially assumed to be identical:

$$V_{hi} = \frac{V_h}{n} \quad (1)$$

$$F_i = \sqrt[n]{1 - ER} \quad (2)$$

$$ER_i = 1 - F_i \quad (3)$$

$$f_{ub}CL_{int,i} = \frac{Q_h ER_i}{1 - ER_i} \quad (4)$$

$$f_{ub}CL_{int} = n f_{ub}CL_{int,i} \quad (5)$$

where Q_h is hepatic blood flow and f_{ub} is the unbound fraction in blood.

According to eqs. 2–5, the total hepatic ER for the SCM with n liver subcompartments is expressed as:

$$ER = 1 - \left(\frac{Q_h}{Q_h + \frac{f_{ub}CL_{int}}{n}} \right)^n \quad (6)$$

The outflow blood concentration from the liver subcompartment i ($C_{out,i}$, $i = 1, 2, 3 \dots n$) is the input concentration for the subsequent subcompartment $i+1$ and is assumed to be in equilibrium with the liver tissue concentration throughout the i th compartment (C_{hi} , $i = 1, 2, 3 \dots n$):

$$K_{phi} = \frac{R_b C_{hi}}{C_{out,i}} \quad (7)$$

where K_{phi} is the liver-to-plasma concentration ratio for liver subcompartment i and R_b is the blood-to-plasma ratio.

It is assumed that each liver subcompartment shares the same K_{phi} for all model compounds except CQ and QD, for which nonlinear binding in the liver has been reported (Harashima et al., 1985; Liu and Jusko, 2021), thus resulting in a concentration-dependent K_{phi} . For the i th ($i = 1, 2, 3 \dots n$) subcompartment, the mathematical relationship between the bound (C_{bhi}) and free (C_{uhi}) hepatic concentration of CQ or QD is expressed by:

$$C_{bhi} = \frac{B_{max} C_{uhi}}{K_d + C_{uhi}} \quad (8)$$

where B_{max} and K_d are the binding capacity of the liver and the equilibrium dissociation constant that were assumed to be the same for each of the subcompartments.

TABLE 1
PK data included in the current analysis

Compound	Dosing Route	Dose (mg/kg)	PK Sample	Literature Sources
Tolbutamide (TB)	i.v.	80	plasma, liver	(Sugita et al., 1982)
	oral	50	plasma	(Nishimura et al., 1998)
Chloroquine (CQ)	i.p.	10	plasma, liver	(Adelusi and Salako, 1982)
	oral	10	plasma	(Parson et al., 2012)
Cyclosporine A (CyA)	i.v. infusion	5.9 (duration: 2 min)	blood, liver	(Kawai et al., 1998)
	oral	5, 10	serum	(Wassef et al., 1985)
Ethoxybenzamide (EB)	i.v.	20, 50, 100	plasma, liver	(Lin et al., 1978)
Fingolimod (FTY720)	i.v.	0.3, 2,	blood, liver	(Meno-Tetang et al., 2006)
	oral	2.8, 7.5	blood, liver	
Nicotine (NIC)	i.v.	0.1	plasma, liver	(Plowchalk et al., 1992)
	i.v. and oral	1	plasma	(Jung et al., 1999)
Pentazocine (PTZ)	i.v. infusion	2 (duration: 2 min)	blood, liver	(Fujio et al., 1983)
	oral	5	plasma	(Khan et al., 2022)
Hexobarbital (HB)	i.v.	60	plasma, liver	(Igari et al., 1982a)
	oral	25	blood	(Van der Graaff et al., 1985)
Phenytoin (DPH)	i.v.	10	serum, liver	(Itoh et al., 1988)
	i.v.	10, 25, 40	blood	(Gerber et al., 1971)
Quinidine (QD)	i.v.	30	plasma, liver	(Harashima et al., 1985)
	oral	10	plasma	(Mori et al., 2012)
Propranolol (PPN)	i.v.	7.5	blood, liver	(Schneck et al., 1977)
	oral	10	plasma	(Shibasaki et al., 1989)
Diazepam (DZP)	i.v.	1.2	plasma, liver	(Igari et al., 1982b)
	oral	10	plasma	(Zhou et al., 2020)
	intraportal	1.2	plasma	(Igari et al., 1984)
Verapamil (VEM)	i.v.	5	plasma, liver	(Yamano et al., 2000)
	oral	10	plasma	(Lee et al., 2012)
	i.v.	10	plasma	
Diltiazem (DLZ)	i.v.	5	plasma, liver	(Yamano et al., 2000)
	oral	20	plasma	(Tsui et al., 1994)

The C_{hi} and C_{uhi} of CQ or QD can be calculated from:

$$C_{hi} = \frac{B_{max} C_{uhi}}{K_d + C_{uhi}} + C_{uhi} \quad (9)$$

and

$$C_{uhi} = f_{uh} C_{hi} = f_{up} \frac{C_{hi}}{K_{phi}} \quad (10)$$

where f_{uh} and f_{up} are the unbound fractions of CQ or QD in the liver and plasma.

Combining eqs. 9 and 10 yields the nonlinear K_{ph} for CQ and QD in the i th ($i = 1, 2, 3 \dots n$) subcompartment:

$$K_{phi} = f_{up} \frac{-(C_{hi} - B_{max} - K_d) + \sqrt{(C_{hi} - B_{max} - K_d)^2 + 4K_d C_{hi}}}{2K_d} \quad (11)$$

For the SCM shown in Fig. 1, the concentration changes in the first and i th ($i = 2, 3 \dots n$) well stirred liver compartment are described by:

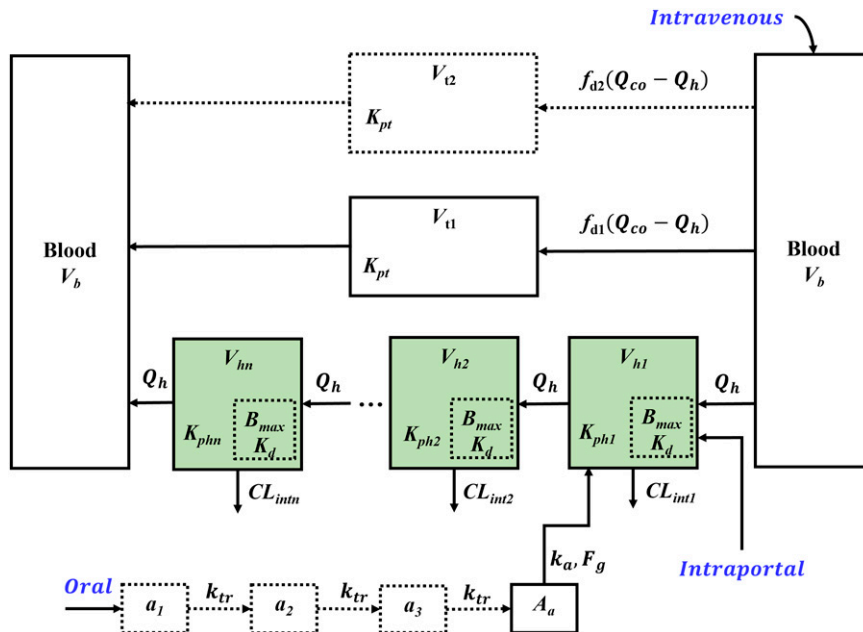


Fig. 1. Minimal PBPK model extended with the hepatic series-compartment model (SCM) containing n liver subcompartments. The model components shown with broken lines are applied as needed.

$$V_{h1} \frac{dC_{h1}}{dt} = Input + Q_h \left(C_b - \frac{R_b C_{h1}}{K_{ph1}} \right) - f_{ub} CL_{int1} \frac{R_b C_{h1}}{K_{ph1}} \quad C_{h1}(0) = 0 \quad (12)$$

and

$$V_{hi} \frac{dC_{hi}}{dt} = Q_h \left(\frac{R_b C_{h(i-1)}}{K_{ph(i-1)}} - \frac{R_b C_{hi}}{K_{phi}} \right) - f_{ub} CL_{inti} \frac{R_b C_{hi}}{K_{phi}} \quad C_{hi}(0) = 0 \quad (13)$$

where C_b is the total concentration in blood and $Input$ is the dosing input for the oral ($Input_{po}$) or intraportal ($Input_{intraportal}$) administration that directly enters the first hepatic compartment (V_{h1}):

$$Input_{po} = k_a F_g A_a$$

$$Input_{intraportal} = Dose_{intraportal}$$

where k_a is the first-order absorption rate constant, F_g is the prehepatic bioavailability, $Dose_{intraportal}$ is the intraportal dose (DZP only) and A_a is the amount of drug in the assumed absorption compartment for all compounds except CyA and PTZ:

$$\frac{dA_a}{dt} = -k_a A_a \quad A_a(0) = Dose_{po} \quad (14)$$

where $Dose_{po}$ is the oral dose.

Delays in oral absorption were observed for both CyA and PTZ, which were best described by a three-transit compartment model (Fig. 1) where the oral dose enters the first transit compartment instead of the absorption compartment. The changes in drug amounts in the transit and absorption compartments are:

$$\frac{da_1}{dt} = -k_{tr} a_1 \quad a_1(0) = Dose_{po} \quad (15)$$

$$\frac{da_2}{dt} = k_{tr}(a_1 - a_2) \quad a_2(0) = 0 \quad (16)$$

$$\frac{da_3}{dt} = k_{tr}(a_2 - a_3) \quad a_3(0) = 0 \quad (17)$$

$$\frac{dA_a}{dt} = k_{tr} a_3 - k_a A_a \quad A_a(0) = 0 \quad (18)$$

where a_1 , a_2 , and a_3 are drug amounts in the first, second, and third transit compartments and k_{tr} is the first-order transit rate constant.

The total liver concentration is assumed to be the average of C_{hi} in each of the subcompartments:

$$C_h = \frac{1}{n} \sum_{i=1}^n C_{hi} \quad (19)$$

The concentration changes in blood (V_b) and two tissue compartments (V_{t1} and V_{t2}) are described by (Cao and Jusko, 2012):

$$V_b \frac{dC_b}{dt} = f_{d1}(Q_{co} - Q_h) \left(\frac{R_b C_{t1}}{K_{pt}} - C_b \right) + f_{d2}(Q_{co} - Q_h) \left(\frac{R_b C_{t2}}{K_{pt}} - C_b \right) + Q_h \left(\frac{R_b C_{hn}}{K_{phn}} - C_b \right) \quad C_b(0) = \frac{Dose_{iv}}{V_b} \quad (20)$$

and

$$V_{t1} \frac{dC_{t1}}{dt} = f_{d1}(Q_{co} - Q_h) \left(C_b - \frac{R_b C_{t1}}{K_{pt}} \right) \quad C_{t1}(0) = 0 \quad (21)$$

and

$$V_{t2} \frac{dC_{t2}}{dt} = f_{d2}(Q_{co} - Q_h) \left(C_b - \frac{R_b C_{t2}}{K_{pt}} \right) \quad C_{t2}(0) = 0 \quad (22)$$

where V_b is blood volume, V_{t1} and V_{t2} are the volumes and C_{t1} and C_{t2}

are the concentrations for tissue compartments 1 and 2, K_{pt} is the shared partition coefficient for the tissue compartments, Q_{co} is cardiac output, f_{d1} and f_{d2} are the distribution fractions of $(Q_{co} - Q_h)$ for V_{t1} and V_{t2} , C_{hn} and K_{phn} are the concentration and partition coefficient for the last liver subcompartment, and $Dose_{iv}$ is the intravenous dose. Given the nonlinear tissue binding of CQ and QD, the K_{phi} ($i = 1, 2, 3 \dots n$) terms in eqs. 12, 13, and 20 were replaced by eq. 11.

Physiologic model parameters were fixed to the literature values listed in Supplemental Table 1 with the following restrictions:

$$f_{d1} + f_{d2} \leq 1 \quad \text{and} \quad V_b + V_h + V_{t1} + V_{t2} = BW(\text{Body Weight})$$

The above model equations with or without the components colored in red apply to the two-tissue or one-tissue mPBPK model extended with the SCM.

Data Analysis. The intravenous blood/plasma and liver concentration-time data in rats for all 14 compounds (Table 1) have been assessed previously (Li and Jusko, 2022) using a piece-wise open-loop approach where the blood PK data were fitted first and then fixed as the forcing input function to model the liver PK data. The major limitation of such an approach is that CL_{int} and K_{ph} are highly correlated, resulting in extremely large CV% values when estimated simultaneously. Therefore, it was necessary to back calculate the CL_{int} from the hepatic ER that was then fixed in the model to only estimate K_{ph} . In this work, the same sets of intravenous PK data were revisited and jointly fitted, applying the one-tissue mPBPK model extended with the hepatic SCM (i.e., eqs. 12, 13, and 19–21 without the components in red). As was done in our most recent work (Li and Jusko, 2023), the SCM containing one, two, and five liver subcompartments was selected herein to mathematically mimic the WSM and the DM with $D_N = 0.6$ and 0.1 , the upper and lower boundaries of the commonly reported D_N range for the liver (Díaz-García et al., 1992; Chou et al., 1993; Evans et al., 1993; Oliver et al., 2001). The CL_{int} and K_{ph} were simultaneously estimated with the hepatic ER being obtained as a secondary parameter using eq. 6. The apparent K_{ph} values ($K_{ph,AR}$) representing the ratio of model-fitted area under the curves of liver and blood concentration-time data (Gallo et al., 1987) were also calculated. The ER , $K_{ph,AR}$, CL_{int} , and K_{ph} estimated in the current modeling analysis were compared with those obtained by the open-loop approach (Li and Jusko, 2022).

Dose-dependent PK was reported for DPH due to the saturable aromatic hydroxylation by CYP2C11 in rat and CYP2C9 in human (Gerber et al., 1971; Doecke et al., 1991; Soucek and Gut, 1992; Ashforth et al., 1995). Accordingly, the blood PK data for DPH after three different i.v. doses (10, 25, and 40 mg/kg) from Gerber et al. (1971) along with the blood and liver PK data after a 10 mg/kg i.v. dose from Itoh et al. (1988) were simultaneously fitted by the extended mPBPK model with the incorporation of saturable metabolism in the liver to quantitatively describe the nonlinear PK characteristics of DPH. The concentration-dependent CL_{int} for DPH is described by the Michaelis-Menten equation:

$$CL_{int} = \frac{V_{max}}{K_m + f_{up} \frac{C_h}{K_{ph}}} \quad (23)$$

where V_{max} is the maximal rate of metabolism, K_m is the Michaelis-Menten constant, and $f_{up} C_h / K_{ph}$ is the unbound hepatic concentration of DPH at the site of metabolism, assuming an instant equilibrium existing between the plasma and liver tissue. Therefore, the concentration changes of DPH in the liver with the consideration of nonlinear hepatic metabolism are expressed as:

$$V_h \frac{dC_h}{dt} = Q_h \left(C_b - \frac{R_b C_h}{K_{ph}} \right) - \frac{V_{max}}{K_m + f_{up} \frac{C_h}{K_{ph}}} f_{up} \frac{C_h}{K_{ph}} \quad C_h(0) = 0 \quad (24)$$

The extended mPBPK model with one hepatic compartment (Fig. 1) was also applied to assess the oral absorption and first-pass effects by

simultaneously characterizing the available intravenous and oral blood and/or liver PK data in rats for all model compounds except EB and DPH (Table 1). Oral PK data for EB could not be found in the literature. Although oral PK data for DPH was available, they were excluded due to data inconsistency. Therefore, the intravenous and oral PK data for 12 out of 14 compounds were included in this part of the analysis (Table 1).

The impacts of n ($n = 1, 2$, and 5) on the model fittings and parameter estimates were also explored using the data for NIC, DZP, VEM, and DLZ. The oral bioavailability (F_{po}) for all compounds were obtained using two methods:

$$F_{po} = \begin{cases} \frac{AUC_{po}Dose_{iv}}{AUC_{iv}Dose_{po}} \\ F_g(1 - ER) \end{cases} \quad (25)$$

where AUC_{iv} and AUC_{po} are the model-fitted area under the curve (AUC) values for the blood concentration-time profile after intravenous and oral dosing.

To further explore the impacts of changing K_{ph} or k_a and metabolic enzyme zonation on intravenous and oral PK profiles in the blood, liver, and each of the liver subcompartments, model simulations were performed for NIC, DLZ, and DPH using the mPBPK model extended with the SCM under the following conditions: 1) different K_{ph} values (i.e., K_{ph} , ± 2 -fold K_{ph} , and ± 5 -fold K_{ph}); 2) different k_a values (i.e., k_a , ± 2 -fold k_a , and ± 5 -fold k_a); and 3) different patterns of hepatic enzyme zonation as were assumed previously (Li and Jusko, 2023) [i.e., either lower metabolic activity at the periportal (PP) region assuming $F_{i+1} = F_i^2$ ($i = 1, 2 \dots 4$);

$$F_1 = \sqrt[1+2+4+8+16]{(1 - ER)} \quad (26)$$

$$F_{i+1} = F_i^2, \quad i = 1, 2 \dots 4 \quad (27)$$

or lower metabolic activity at the perivenous (PV) region assuming $F_{i+1} = F_i^{1/2}$ ($i = 1, 2 \dots 4$):

$$F_1 = \sqrt[1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}]{(1 - ER)} \quad (28)$$

$$F_{i+1} = F_i^{\frac{1}{2}}, \quad i = 1, 2 \dots 4 \quad (29)$$

With the assumed F_i values, the corresponding $f_{ub}CL_{int}$ were obtained using eqs. 3 and 4 for the subsequent model simulations. The PK parameters including C_{max} , time to reach C_{max} (T_{max}), and AUC of the simulated data were obtained by noncompartmental analysis (NCA) performed using Phoenix WinNonlin version 6.4 (Certara USA, Inc., Princeton, NJ), and the fold changes in these parameters relative to the nominal values obtained at $1 \times K_{ph}$, $1 \times k_a$ or even enzyme distribution were assessed.

The model-fitted fractional distribution parameter f_d for all of the compounds was examined in relation to apparent parallel artificial membrane permeability assay (PAMPA) permeability coefficient (P) values (i.e., $f_{up}P/R_b$). Compound-specific physicochemical properties and blood partitioning parameters used for calculating the apparent PAMPA P were listed in Supplemental Table 3.

Model Fitting. The model fittings of blood and liver concentration-time data were performed by nonlinear regression using the maximum likelihood algorithm in ADAPT 5 (Biomedical Simulations Resources, Los Angeles, CA) (D'Argenio et al., 2009). The variance model was:

$$V_i = (\sigma_{inter} + \sigma_{slope}Y_i)^2 \quad (30)$$

where V_i is the variance of the i th data point, Y_i is the i th model-predicted concentration, and σ_{inter} and σ_{slope} are the variance model parameters. Model selection was based on the goodness-of-fit criteria, which

included the Akaike information criterion (AIC), visual inspection of the fitted profiles, and CV% of the parameter estimates.

The ADAPT code for 1) one-tissue mPBPK extended with the SCM ($n = 5$) and nonlinear tissue binding for the intravenous PK of QD; 2) two-tissue mPBPK extended with the SCM ($n = 5$) for the intravenous, oral, and intraportal PK of DZP; 3) one-tissue mPBPK with absorption delays for the intravenous and oral PK of CyA; and 4) one-tissue mPBPK with nonlinear hepatic metabolism for the intravenous PK of DPH are provided in the Supplemental Material.

Results

Assessing Hepatic Distribution and Clearance by the SCM: mPBPK Model versus Open-Loop Approach. The reported blood and liver concentration-time data for each of the 14 compounds after intravenous administration (Table 1) were simultaneously described by the one-tissue mPBPK model extended with the hepatic SCM ($n = 1, 2$, and 5) shown in Fig. 1. The model fittings are presented in Fig. 2, and the final parameter estimates are listed in Table 2. As can be seen, the observed data for all compounds were well fitted with low CV% values for most parameter estimates except for the K_{ph} and CL_{int} values of PPN and DZP when fitted by the one-compartment SCM (i.e., WSM), indicating that these parameters were not well estimated by the WSM given the currently available PK data. The nonlinear K_d values for CQ and QD tend to have higher CV%, which may be attributed to the limited range of concentrations producing some difficulty in fitting a nonlinear function. The early phases of the liver profiles for CyA and EB that were previously underpredicted by the open-loop approach (Li and Jusko, 2022) were nicely captured by the extended mPBPK model. For CQ and QD, the nonparallel terminal phases of blood and liver PK profiles were reasonably explained by incorporating the nonlinear tissue binding in the hepatic compartments. The reported intravenous PK profiles of CQ were obtained after an intraperitoneal (i.p.) dose but were better fitted without an absorption phase as presented given the very fast absorption and low first-pass extraction of CQ (Al Shoyaib et al., 2019). The B_{max} and K_d estimates for CQ are close to those obtained in our previous analysis ($B_{max} = 12,000$ ng/ml; $K_d = 8$ ng/ml) (Li and Jusko, 2022).

As was shown, the one-tissue mPBPK model with the SCM containing different n yielded mostly comparable values of the hepatic ER and $K_{ph,AR}$ for each of the compounds. They exhibited similar model performance as indicated by the AIC values (Table 2) and almost overlapping model fittings with slight differences in the early phases (Fig. 2); however, there is a tendency that the CV% of CL_{int} and K_{ph} for some of the compounds is larger with $n = 1$, suggesting that the fittings are more reliable with a larger n . The CL_{int} and K_{ph} estimates for each of the compounds decrease with increasing n and such dependency becomes more remarkable as ER increases, consistent with previous assessments applying the open-loop approach (Li and Jusko, 2022).

As indicated in Fig. 3, the estimated ER and $K_{ph,AR}$ by the mPBPK model with the SCM are independent of the number of liver subcompartments and correlate well with those obtained by the open-loop approach within a 2-fold error (Fig. 3, A and B). According to the quantitative correlations between the SCM and the basic hepatic clearance models demonstrated previously (Li and Jusko, 2023), the comparisons of CL_{int} and K_{ph} were made between the SCM with $n = 1$ (i.e., mPBPK SCM-1) and the WSM and between the SCM with $n = 2$ (i.e., mPBPK SCM-2) and the DM with $D_N = 0.6$ (Fig. 3, C–F). As shown, the two different modeling approaches gave comparable values of CL_{int} and K_{ph} for most of the compounds except for PPN and DZP. Comparing to

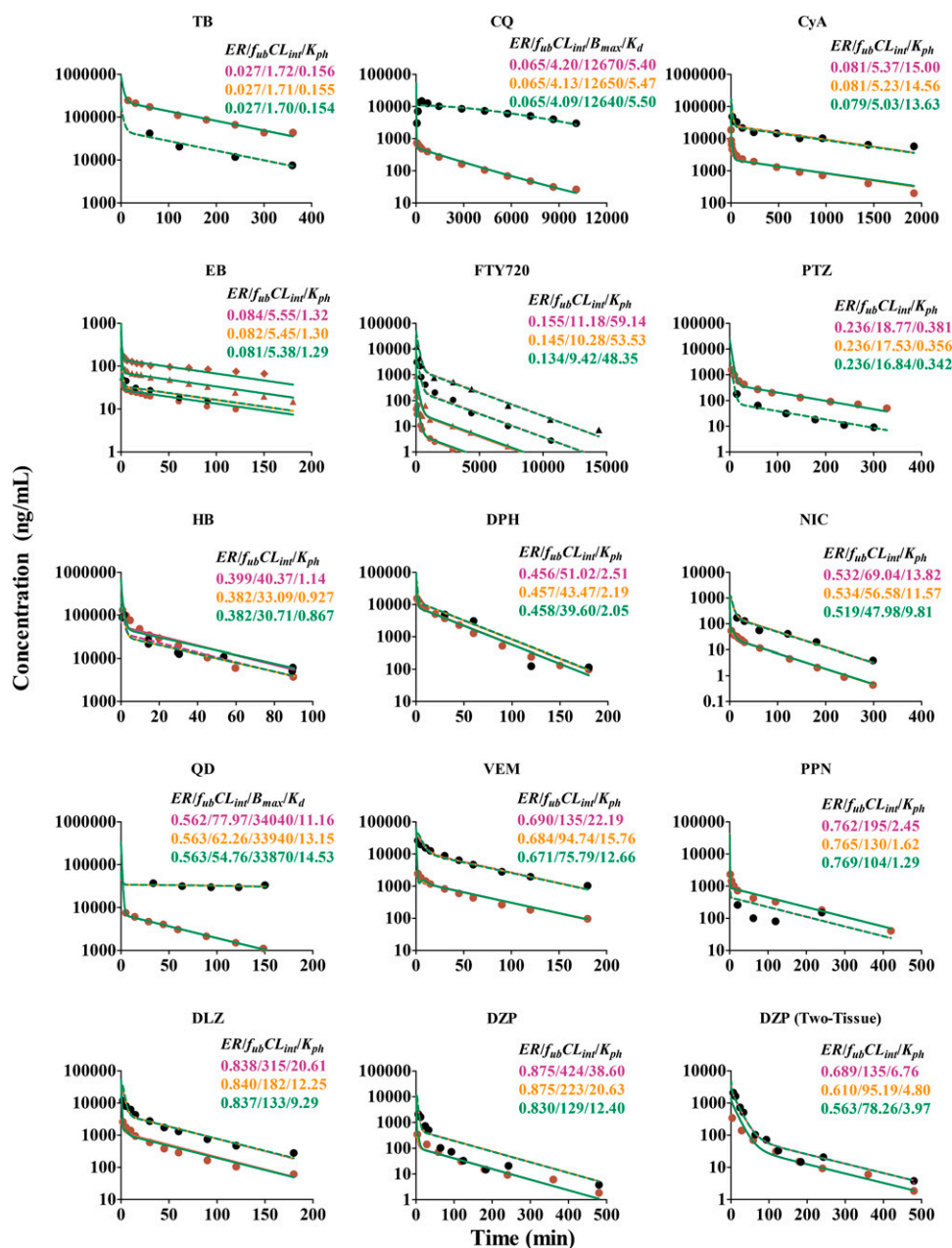


Fig. 2. Blood and liver concentration-time profiles of 14 model compounds after intravenous dosing. Measured concentrations in blood and liver are indicated by solid symbols in red and black. Solid and dashed lines represent model fittings of blood and liver data by one-tissue mPBPK extended with the SCM containing one (purple), two (orange), and five (green) liver subcompartments. The estimated ER , $f_{ub}CL_{int}$, and K_{ph} (B_{max} and K_d for CQ and QD) values for each of the compounds are listed with the same color coding as those for the model fittings.

the open-loop approach, the mPBPK model slightly overestimated the CL_{int} of PPN by 2.26-fold with SCM-1 (Fig. 3C); nonetheless, the resulting ER (0.76) is comparable to the reported value (0.77) (Li and Jusko, 2022). In addition, the CL_{int} and K_{ph} of DZP were overestimated by the one-tissue mPBPK model extended with the SCM by 2.48~3.59-fold (Fig. 3, C and E) and 2.85~4.05-fold (Fig. 3, D and F), with a larger n yielding a smaller deviation from the open-loop approach. Such overestimation may be related to the overprediction at the terminal phase of the liver PK profile for DZP. Therefore, a two-tissue mPBPK model with the SCM was further applied to the PK data for DZP, and the fitting results are indicated as DZP (two-tissue) in Fig. 2. As can be seen, the intravenous liver PK for DZP was better captured by the two-tissue

model and the estimated parameters are more consistent with those from the open-loop approach (Fig. 3).

Assessing Nonlinear Metabolism. The nonlinear PK characteristics of DPH were reasonably described by incorporating saturable metabolism in the hepatic compartment of the extended mPBPK model (Fig. 4), yielding parameters with low CV% values (Table 3). Previously, a two-site Michaelis-Menten model consisting of a saturable high-affinity, low-capacity site and a nonsaturable low-affinity, high-capacity site was applied to the intravenous infusion data for DPH with a steady-state (SS) plasma concentration (C_p) range of 378.4~22542 ng/ml, which generated a V_{max} of 70,560 ng/min/kg and a K_m of 504 ng/ml based on the unbound DPH concentration (Ashforth et al., 1995). The estimated V_{max} in the present analysis

TABLE 2

Final parameter estimates for 14 model compounds obtained by jointly fitting the blood and liver concentration-time data after single intravenous dosing by the mPBPK model extended with the hepatic SCM ($n = 1, 2$, and 5)

Compound	SCM	Estimates (CV%)						AIC
		ER	f_d	K_{pt}	K_{ph}	$f_{ub}CL_{int}$ (ml/min/kg)	$K_{ph,AR}^a$	
Tolbutamide	$n = 1$	0.027 (4.27)	0.045 (38.90)	0.183 (8.73)	0.156 (7.63)	1.72 (4.39)	0.152	256
	$n = 2$	0.027 (4.27)	0.045 (38.76)	0.183 (8.72)	0.155 (7.61)	1.71 (4.36)	0.152	256
	$n = 5$	0.027 (4.28)	0.045 (39.12)	0.183 (8.74)	0.154 (7.62)	1.70 (4.35)	0.152	256
Chloroquine	$n = 1$	0.065 (7.51)	0.021 (27.55)	38.18 (16.10)	B_{max} : 12670 (11.10) K_d : 5.40 (28.56) K_{ph} : 2.92~718 ^b	4.20 (8.03)	149.6	345
	$n = 2$	0.065 (7.51)	0.021 (27.53)	38.14 (16.10)	B_{max} : 12650 (11.08) K_d : 5.47 (28.56) K_{ph} : 2.89~708 ^b	4.13 (7.89)	149.7	345
	$n = 5$	0.065 (7.53)	0.021 (27.66)	38.19 (16.16)	B_{max} : 12640 (11.06) K_d : 5.50 (28.54) K_{ph} : 2.88~703 ^b	4.09 (7.84)	149.9	345
Cyclosporine A	$n = 1$	0.081 (7.87)	0.211 (14.07)	5.70 (10.81)	15.00 (9.91)	5.37 (8.56)	13.77	409
	$n = 2$	0.081 (7.81)	0.204 (14.12)	5.69 (10.84)	14.56 (9.39)	5.23 (8.32)	13.66	409
	$n = 5$	0.079 (8.57)	0.168 (14.72)	5.66 (12.06)	13.63 (9.69)	5.03 (9.00)	12.97	414
Ethoxybenzamide	$n = 1$	0.084 (4.57)	0.397 (13.87)	0.624 (4.04)	1.32 (6.05)	5.55 (4.99)	1.21	283
	$n = 2$	0.082 (4.45)	0.424 (14.49)	0.622 (4.01)	1.30 (6.06)	5.45 (4.85)	1.21	283
	$n = 5$	0.081 (4.37)	0.427 (14.07)	0.622 (4.00)	1.29 (6.07)	5.38 (4.76)	1.22	283
Fingolimod	$n = 1$	0.155 (8.62)	0.016 (18.47)	7.51 (12.99)	59.14 (11.54)	11.18 (10.20)	34.10	396
	$n = 2$	0.145 (8.38)	0.016 (18.60)	7.20 (13.12)	53.53 (10.86)	10.28 (9.80)	32.53	399
	$n = 5$	0.134 (7.41)	0.014 (18.06)	6.76 (12.67)	48.35 (8.66)	9.42 (8.56)	30.79	401
Pentazocine	$n = 1$	0.236 (6.42)	0.034 (15.85)	1.29 (12.74)	0.381 (15.40)	18.77 (8.40)	0.291	169
	$n = 2$	0.236 (6.44)	0.034 (15.85)	1.30 (12.74)	0.356 (15.07)	17.53 (7.89)	0.291	169
	$n = 5$	0.236 (6.45)	0.034 (15.86)	1.30 (12.75)	0.342 (14.90)	16.84 (7.60)	0.292	169
Hexobarbital	$n = 1$	0.399 (9.87)	0.475 (23.44)	0.827 (14.04)	1.14 (18.65)	40.37 (16.41)	0.686	408
	$n = 2$	0.382 (9.61)	0.309 (22.63)	0.809 (14.31)	0.927 (16.45)	33.09 (13.88)	0.650	408
	$n = 5$	0.382 (9.47)	0.293 (22.24)	0.805 (14.29)	0.867 (15.42)	30.71 (12.75)	0.655	408
Phenytoin	$n = 1$	0.456 (8.52)	0.634 (29.74)	0.895 (11.87)	2.51 (20.35)	51.02 (15.67)	1.37	252
	$n = 2$	0.457 (8.43)	0.586 (29.86)	0.885 (11.84)	2.19 (19.07)	43.47 (13.49)	1.40	252
	$n = 5$	0.458 (8.28)	0.509 (29.41)	0.869 (11.75)	2.05 (18.27)	39.60 (12.13)	1.44	252
Nicotine	$n = 1$	0.532 (4.09)	0.212 (17.72)	1.60 (7.76)	13.82 (9.73)	69.04 (8.72)	6.47	89
	$n = 2$	0.534 (3.77)	0.199 (17.31)	1.56 (7.45)	11.57 (8.10)	56.58 (6.80)	6.64	87
	$n = 5$	0.519 (3.58)	0.176 (17.27)	1.48 (7.51)	9.81 (6.73)	47.98 (5.67)	6.46	88
Quinidine	$n = 1$	0.562 (3.00)	0.202 (11.24)	2.79 (5.56)	B_{max} : 34040 (6.13) K_d : 11.16 (96.03) K_{ph} : 9.03~120 ^b	77.97 (6.86)	57.45	207
	$n = 2$	0.563 (2.99)	0.205 (11.23)	2.80 (5.53)	B_{max} : 33940 (6.10) K_d : 13.15 (97.47) K_{ph} : 8.24~100 ^b	62.26 (5.68)	65.22	207
	$n = 5$	0.563 (2.99)	0.207 (11.26)	2.81 (5.51)	B_{max} : 33870 (6.10) K_d : 14.53 (99.22) K_{ph} : 7.77~89.33 ^b	54.76 (5.04)	72.14	207
Verapamil	$n = 1$	0.690 (4.23)	0.447 (12.26)	1.88 (6.62)	22.19 (14.12)	135.1 (13.63)	6.85	300
	$n = 2$	0.684 (4.38)	0.395 (12.57)	1.82 (7.10)	15.76 (11.45)	94.74 (10.81)	6.88	302
	$n = 5$	0.671 (4.64)	0.359 (13.34)	1.75 (7.8)	12.66 (10.09)	75.79 (9.51)	6.79	306
Propranolol	$n = 1$	0.793 (11.41)	0.884 (18.02)	7.63 (13.68)	2.81 (57.83)	233 (55.13)	0.58	178
	$n = 2$	0.801 (11.40)	0.901 (18.63)	7.70 (13.70)	1.83 (45.31)	151 (41.5)	0.59	178
	$n = 5$	0.806 (11.51)	0.931 (19.37)	7.76 (13.84)	1.42 (38.79)	118 (34.19)	0.59	178
Diltiazem	$n = 1$	0.838 (6.59)	0.405 (17.60)	2.02 (10.16)	20.61 (40.55)	315 (40.73)	3.33	295
	$n = 2$	0.840 (6.33)	0.366 (17.08)	1.97 (10.12)	12.25 (27.60)	182 (27.64)	3.43	295
	$n = 5$	0.837 (5.97)	0.322 (16.40)	1.90 (10.09)	9.29 (20.21)	133 (20.18)	3.55	295
Diazepam	$n = 1$	0.875 (9.86)	0.183 (24.56)	3.13 (15.25)	38.60 (76.97)	424 (78.61)	4.84	195
	$n = 2$	0.875 (9.59)	0.176 (25.16)	3.08 (15.66)	20.63 (49.86)	223 (52.06)	4.93	195
	$n = 5$	0.830 (9.71)	0.151 (25.45)	2.76 (16.50)	12.40 (29.32)	129 (31.79)	4.84	195
Diazepam ^c (Two-Tissue)	$n = 1$	0.689	0.999	2.23	6.76	135	2.10	167
	$n = 2$	0.610	1.000	2.20	4.80	95.19	2.10	168
	$n = 5$	0.563	0.999	2.17	3.97	78.26	2.10	168

^a $K_{p,AR}$ is the apparent K_p calculated by the area ratio (AR) method (Gallo et al., 1987).

^bThe K_{ph} ranges of CQ and QD were obtained from the estimated B_{max} and K_d with the C_h range of 2.98~14.64 $\mu\text{g/ml}$ for CQ and 30~35 $\mu\text{g/ml}$ for QD, which covers the measured liver concentrations in the source datasets.

^cThe f_d values for DZP estimated by the two-tissue mPBPK model were calculated as the sum of f_{d1} and f_{d2} .

(86,630 ng/min/kg) is comparable, whereas the K_m (91.68 ng/ml) is lower. Due to the saturable metabolism, the ER of DPH decreases with increasing hepatic concentrations and was predicted to be 0.04~0.78 within the predicted liver concentration range of 1~52433 ng/ml according to eqs. 6 and 24.

Assessing Oral Absorption and First-Pass Effects. The oral absorption and first pass of 12 compounds were assessed using the

mPBPK model extended with the SCM. During the preliminary analysis, alternative structures of the mPBPK models were assessed such as using more tissue compartments, the same or different K_{pt} values, and the same or different f_d values for multiple tissue spaces, with or without absorption delays. Finally, the one-tissue mPBPK model extended with the SCM was sufficient for jointly fitting the oral and intravenous data for TB, CyA, CQ, FTY720, HB, NIC, QD, and

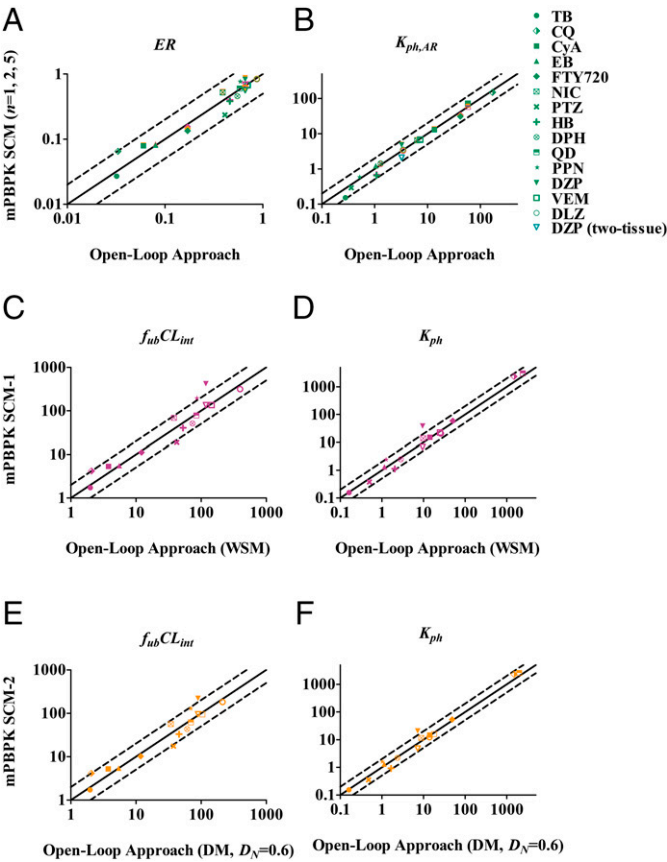


Fig. 3. Comparisons of the estimated ER (A), $K_{ph,AR}$ (B), $f_{ub}CL_{int}$ (C and E), and K_{ph} (D and F) by the one-tissue mPBPK model with the SCM ($n = 1, 2$, and 5) and those obtained by the open-loop approach (Li and Jusko, 2022). The K_{ph} values for CQ and QD shown were calculated from B_{max}/K_{d} . The parameter values for DZP estimated by the two-tissue mPBPK model with the SCM are also plotted for comparisons (open triangles). The solid line represents unity, and the dashed lines indicate 2-fold range from unity. The color coding is the same as used in Fig. 2.

PPN. However, after inclusion of the oral PK data, two-tissue compartments (V_1 and V_2) sharing the same K_{pt} in addition to the hepatic compartment were found to be optimal for DZP, VEM, DLZ, and PTZ. As shown in Figs. 5–7, there is good agreement between the model predictions and observations obtained with the different dosing routes. The fitted parameters for all compounds were precisely estimated with low CV% values except that the modeling software was unable to generate CV% values for DLZ (Table 4).

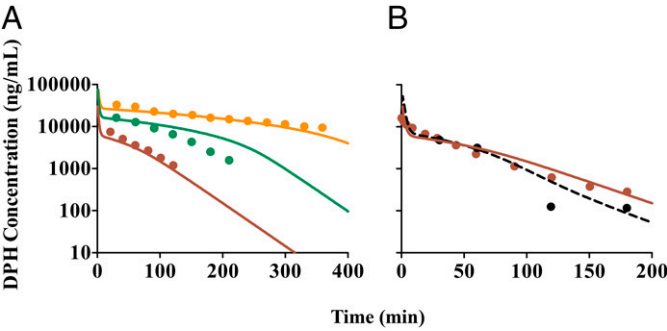


Fig. 4. Blood and liver concentration-time profiles of DPH after i.v. bolus administration of 10- (red), 25- (green), or 40- (orange) mg/kg doses. Closed circles and solid lines represent the observations and model fittings of blood concentrations. Open circles and dashed lines indicate the observed and model-fitted liver concentrations. The measured data in panel (A) were obtained from Gerber et al. (1971), and those in panel (B) were from Itoh et al. (1988).

TABLE 3

Final parameter estimates for DPH obtained by fitting the i.v. blood (10, 25, and 40 mg/kg) and liver (10 mg/kg) PK data using the mPBPK model extended with a liver compartment and saturable hepatic metabolism

Parameter (Unit)	Estimates	CV%
f_d	0.351	21.95
K_{pt}	1.460	8.52
K_{ph}	1.462	21.13
V_{max} (ng/min/kg)	86630	7.84
K_m (ng/ml)	91.68	37.72

For most compounds, parameters related to the hepatic distribution and elimination (hepatic ER , CL_{int} , and K_{ph}) were comparable to those obtained without the oral PK data (Table 2). The oral absorption of all selected compounds except for CyA and PTZ was well described by a first-order process with the estimated absorption rate constant (k_a) ranging from 0.0007 minutes⁻¹ for FTY720 to 0.149 minutes⁻¹ for HB. The oral absorption delays for CyA and PTZ were reasonably captured by incorporating the three-compartment transit model yielding first-order transit rate constants (k_{tr}) of 0.019 and 0.089 minutes⁻¹. The oral dosing vehicle for CyA was olive oil (Wassef et al., 1985), which might account for the delay in absorption by retarding gastric emptying (Ueda et al., 1984). PTZ was also reported to potentially

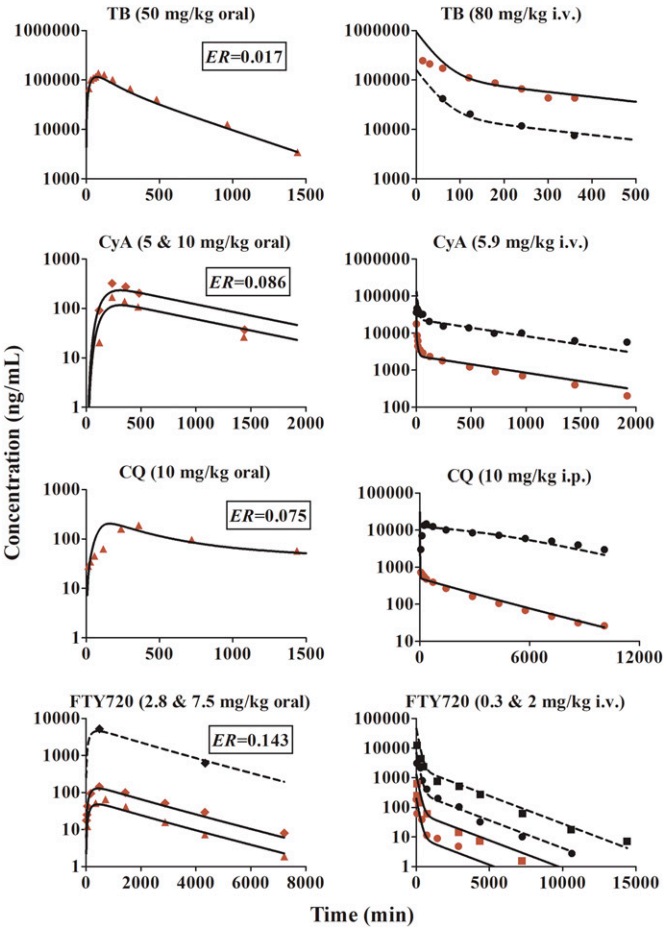


Fig. 5. Blood and liver concentration-time profiles of TB, CyA, CQ, and FTY720 after oral, intravenous, or intraperitoneal (CQ only) dosing. Measured concentrations in blood and liver are indicated by solid and open symbols. Solid and dashed purple lines represent model fittings of blood and liver data by the SCM with one liver subcompartment (=WSM). The estimated ER value for each of the compounds is listed.

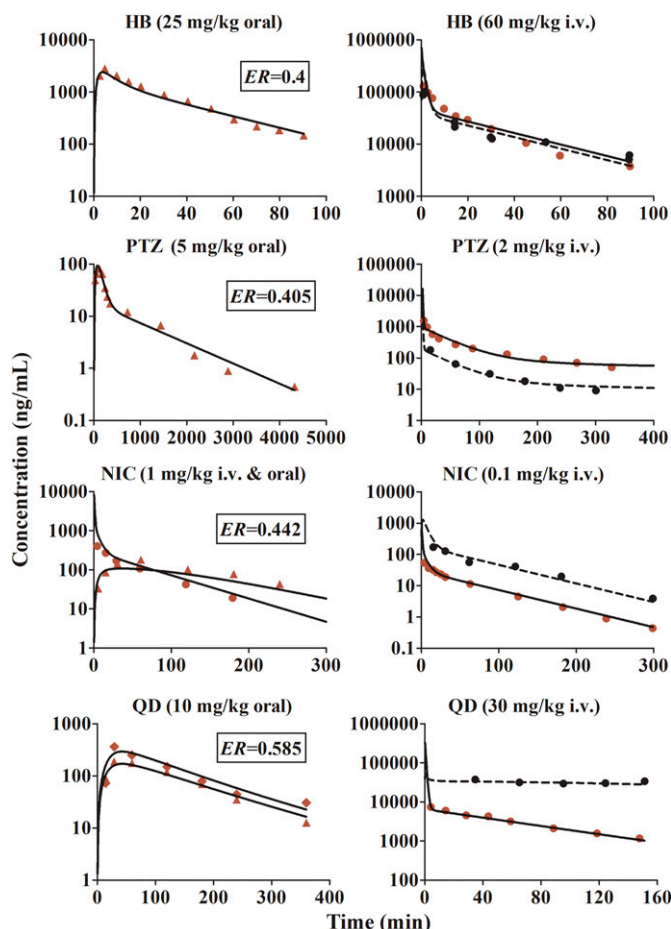


Fig. 6. Blood and liver concentration-time profiles of HB, PTZ, NIC, and QD after oral or intravenous dosing. Symbols, lines, and color coding are the same as used in Fig. 5.

delay gastric emptying time and inhibit propulsive activity in the rat (Danhof et al., 1966).

For TB, NIC, and DLZ, the prehepatic bioavailability (F_g) was fixed to 1 as the initial estimate exceeded 1, indicating that the prehepatic loss of the oral formulations for these compounds is negligible. In contrast, the oral dosage forms of the remaining compounds exhibit varying degrees of prehepatic loss, with the estimated F_g being the lowest for HB (~ 0.11) and the highest for PPN (~ 0.59). The F_{po} for each of the compounds calculated using eq. 24 from the model estimated F_g and ER and from the ratio of model-fitted AUC values demonstrate good consistencies within a 2-fold range (Supplemental Fig. 1) except for CyA. In the original paper where the oral PK data of CyA were obtained (Wassef et al., 1985), the influences of different dosing routes on the PK of CyA were assessed in rats and the bioavailability of the oral route at 5 and 10 mg/kg relative to the extravascular dosing routes that gave the highest AUC were reported to be 10.3%–13.3%. This is comparable with the F_{po} values for CyA (7% and 14.5%) estimated in the current work. The differing absorption and bioavailability of QD after oral doses as a solution or powder in hydroxypropyl methylcellulose (HPMC) capsule were well discerned by the model by assigning dosage form-specific k_a (0.033 vs. 0.023 min^{-1}) and F_g (0.311 vs. 0.405) values. The oral bioavailability (F_{po}) for the QD powder calculated according to eq. 25 was 31%–36% higher than that for the solution and comparable to that reported (36%) (Mori et al., 2012).

Assessing Fractional Distribution and Partition Coefficients. In mPBPK models, the f_d values may represent either differential blood distribution to tissues/organs and/or flow/permeability limitations provided there is no transporter involvement. The ability to resolve such values offers an advantage over the use of forcing functions. The well characterized liver PK for all compounds is supportive of the assumption that distribution into the liver is flow limited (i.e., $f_d = 1$). The high estimate of f_{d1} for PPN and $f_{d1} + f_{d2}$ for DZP and DLZ indicates that the distribution of these compounds into nonliver tissues is also primarily controlled by blood flow. For the remaining compounds, the f_{d1} or $f_{d1} + f_{d2}$ was estimated to be low, ranging from 0.005 for TB and 0.4 for VEM, suggesting that permeability-limited distribution may be operative for some of the nonliver tissues. Such heterogeneous tissue distribution has been reported for CQ (Liu and Jusko, 2021), HB (Igari et al., 1982a), FTY720 (Meno-Tetang et al., 2006), TB (Sugita et al., 1982), and CyA (Kawai et al., 1998). As shown by Jeong et al. (2017), f_d values can be positively correlated with the apparent PAMPA P (i.e., $f_{up}P/R_b$). The relationship between the tissue f_d values obtained herein and the apparent PAMPA P is in line with such findings (Fig. 8).

The distribution of CyA, FTY720, NIC, QD, VEM, and DLZ into the liver is more extensive than into the other tissues in the body as reflected by the appreciably larger K_{ph} than the K_{pt} , although the accumulation of PTZ and PPN in liver is much lower than in other tissues, which is in line with previous findings (Schneck et al., 1977; Fujio et al., 1983). The estimated K_{ph} and K_{pt} are similar (within 2-fold) for

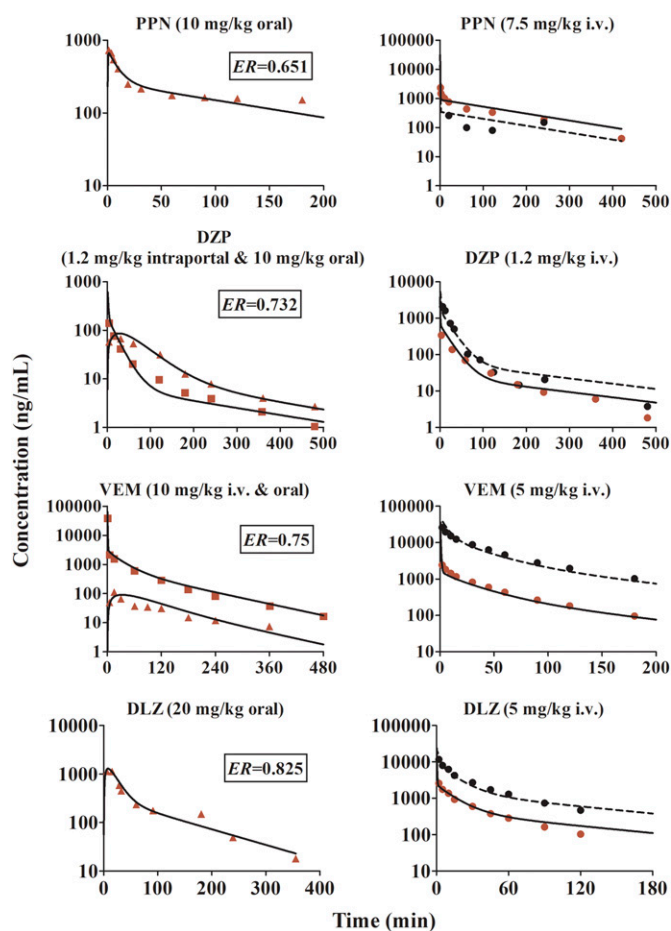


Fig. 7. Blood and liver concentration-time profiles of PPN, DZP, VEM, and DLZ after oral, intravenous, or intraportal (DZP only) dosing. Symbols, lines, and color coding are the same as used in Fig. 5.

TABLE 4

Final parameter estimates for 12 model compounds obtained by jointly fitting the blood and liver concentration-time data after single intravenous, intraperitoneal (CQ only), intraportal (DZP only), and oral dosing with one-/two-tissue mPBPK extended with the hepatic compartment and compound-specific complexities shown in Fig. 1

Parameter Estimates (CV%)										
Compound	ER	$f_{ub}CL_{int}$ (ml/min/kg)	K_{ph}	K_{pt}	f_{d1}	f_{d2}	V_I (ml/kg)	k_a (min ⁻¹)	F_g	k_{tr} (min ⁻¹)
TB	0.017 (4.95)	1.06 (5.04)	0.129 (12.22)	0.18 (12.51)	0.005 (17.49)	—	—	0.009 (12.82)	1 (fixed)	—
CQ	0.075 (11.59)	4.90 (12.52)	B_{max} : 16350 (17.29) K_d : 11.32 (33.26) K_{ph} : 64~473 ^a	58.26 (21.84)	0.042 (45.85)	—	—	0.004 (21.81)	0.157 (23.27)	—
CyA	0.086 (8.47)	5.74 (9.27)	15.81 (11.12)	5.57 (11.44)	0.210 (15.91)	—	—	0.019 (10.42)	0.159 (13.40)	0.019 (10.42)
FTY720	0.143 (7.05)	10.12 (8.22)	55.31 (9.35)	6.70 (11.34)	0.014 (16.11)	—	—	0.0007 (10.43)	0.429 (10.87)	—
NIC	0.442 (4.42)	48.11 (7.92)	10.83 (11.93)	1.06 (10.04)	0.101 (20.17)	—	—	0.011 (17.14)	1 (fixed)	—
HB	0.400 (7.54)	40.46 (12.55)	1.60 (15.45)	0.784 (11.73)	0.185 (19.51)	—	—	0.149 (25.54)	0.114 (14.51)	—
PTZ	0.522 (5.36)	66.47 (11.21)	1.53 (13.55)	34.50 (6.42)	0.229 (18.24)	0.129 (10.28)	67.13 (15.10)	0.012 (19.08)	0.442 (12.75)	0.083 (18.42)
QD	0.585 (3.79)	85.78 (9.14)	B_{max} : 32920 (5.76) K_d : 1.91 (65.35) K_{ph} : 5.39~500 ^a	3.22 (8.62)	0.242 (12.50)	—	—	0.022 (9.37) ^b 0.032 (10.63) ^c	0.442 (8.66) ^b 0.338 (9.02) ^c	—
PPN	0.651 (10.19)	113.7 (29.23)	1.27 (32.39)	8.12 (12.15)	0.959 (14.81)	—	—	0.108 (31.36)	0.587 (28.03)	—
DZP	0.732 (3.43)	166.4 (12.82)	9.13 (14.27)	4.97 (16.43)	0.588 (58.37)	0.055 (15.35)	198.3 (17.28)	0.020 (20.01)	0.181 (17.35)	—
VEM	0.753 (5.01)	185 (20.28)	34.93 (21.01)	2.92 (9.08)	0.326 (18.10)	0.066 (39.53)	392 (22.69)	0.020 (13.81)	0.239 (21.65)	—
DLZ	0.825	286	18.01	4.19	0.83	0.17	233.8	0.122	1 (fixed)	—

^aThe K_{ph} ranges of CQ and QD were obtained as described in Table 2. The estimated K_d of QD after including the oral data are about 5-fold smaller than that obtained with only intravenous data (Table 2), resulting in a much larger K_{ph} range.

^bThe k_a and F_g values are specifically for the oral powder of QD.

^cThe k_a and F_g values are specifically for the oral solution of QD.

—, not applicable.

TB, HB, and DZP, suggesting that the extent of distribution into the liver and nonliver tissues is comparable. Both the K_{ph} (simulated from B_{max} and K_d) and K_{pt} of CQ were estimated as high. Such extensive tissue distribution of CQ is consistent with a previous report (Liu and Jusko, 2021) and is expected, as CQ is subject to lysosomal trapping and binds appreciably to the acidic phospholipids on cell membranes (Allison and Young, 1964; Tietz et al., 1990; Zheng et al., 2011). Non-linear tissue binding of CQ and QD was also reported for nonliver tissues (Harashima et al., 1985; Liu and Jusko, 2021); nevertheless, a constant rather than concentration-dependent K_{pt} representing the time/concentration-averaged value was assigned here for simplicity given the limited data available and was shown to be sufficient for the purpose of the current analysis.

Role of Numbers of Liver Subcompartments. The effects of different numbers of liver subcompartments ($n = 1, 2$, and 5) were further explored using the intravenous and oral PK data for NIC, DZP, VEM, and DLZ, with the model fitting results presented in Supplemental Fig. 2 and Supplemental Table 2. All parameters were precisely estimated except that the software was not able to generate the CV% values for

DZP (when $n = 5$) and DLZ (all models). As expected, CL_{int} and K_p decrease with increasing n , whereas no major difference was observed in the model fittings and estimates for other parameters with varying values of n except that the k_a for VEM was estimated larger (0.099 minutes⁻¹) when $n = 5$, producing a higher peak for the fitted oral blood PK profile.

Assessing Impacts of Changing K_{ph} or k_a and Hepatic Enzyme Zonation on the Blood, Liver, and Intrahepatic Concentrations.

To assess how the intravenous and oral PK profiles are affected by the changes in K_{ph} or k_a , model simulations were performed using the mPBPK model with the SCM ($n = 1$ and 5) utilizing the final parameter estimates for NIC and DLZ listed in Table 4 and varying values of K_{ph} or k_a . The simulated PK profiles with differed K_{ph} are presented in Fig. 9 for DLZ and Supplemental Fig. 3 for NIC, and the fold changes in the resulting PK parameters (i.e., C_{max} , T_{max} , and AUC_{inf}) relative to the corresponding nominal values at $1 \times K_{ph}$ are displayed in Fig. 10 for DLZ and Supplemental Fig. 4 for NIC. No changes were observed in the intravenous blood PK with changing K_{ph} (data not shown), whereas the intravenous liver PK and the PK profiles for blood and liver after oral dosing are all sensitive to the changes in K_{ph} , with the oral liver PK being the most affected. All of the PK parameters exhibit a decreasing trend as K_{ph} becomes smaller except for unchanged AUC_{inf} and slightly increased C_{max} for the oral blood PK profiles. Regardless of the dosing route, the fold changes in AUC_{inf} of the liver PK profiles are always the same as those in K_{ph} . The changes in PK parameters are independent of the number of liver subcompartments except that the T_{max} of the oral blood PK profiles are more sensitive to the change in K_{ph} when $n = 5$.

Figure 11 displays the simulated PK profiles for DLZ in blood, liver, and each of the liver subcompartments after oral dosing with varying k_a . The fold changes in the corresponding T_{max} and C_{max} of the simulated profiles comparing to the nominal values obtained at $1 \times k_a$ are presented in Fig. 12. In line with the intrahepatic concentration gradient demonstrated previously (Li and Jusko, 2023), more delayed T_{max} and smaller C_{max} are observed for the later subcompartments (Fig. 11). All of the PK profiles after oral administration are affected by the change of k_a (i.e., when k_a decreases, the T_{max} increases), with the most

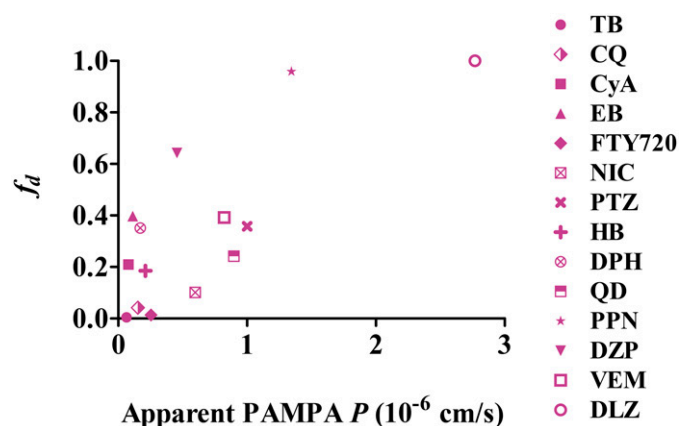


Fig. 8. Relationship between the model estimated tissue f_d and the apparent PAMPA P (i.e., $f_{up}P/R_b$).

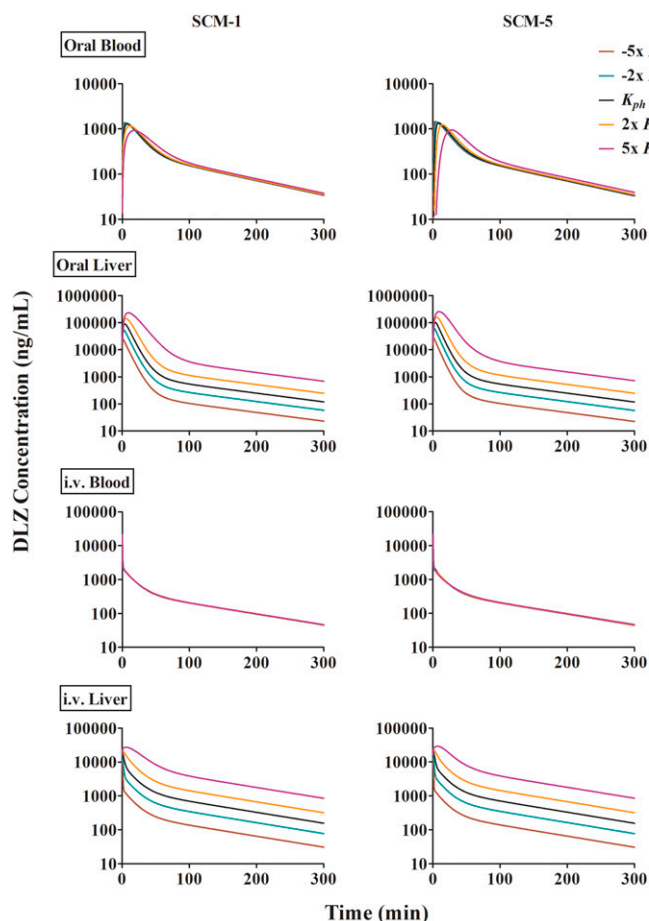


Fig. 9. Effects of changing K_{ph} on predicting the blood and liver concentration-time profiles for DLZ after intravenous or oral dosing by the mPBPK model extended with the SCM ($n = 1$ and 5). Color coded lines represent model simulations with K_{ph} (17.87 for SCM1 and 8.09 for SCM5, black), -2 -fold K_{ph} (green), $+2$ -fold K_{ph} (orange), -5 -fold K_{ph} (red), and $+5$ -fold K_{ph} (purple).

remarkable changes observed for the blood PK profiles especially with the smaller k_a values, whereas the C_{max} decreases with that for the liver PK exhibiting a steeper decline than that for the blood PK. The intrahepatic PK profiles were affected differently by the alteration of k_a , with the first subcompartment being the most sensitive followed by each of the subsequent compartments. The AUC_{inf} for all PK profiles remained unchanged with varying k_a (data not shown).

The impacts of metabolic zonation on the intravenous and oral PK were assessed using the SCM with five liver subcompartments assuming a) lower metabolic activity at the PP region, b) uniform metabolic activity, or c) lower metabolic activity at the PV region, where each of the liver subcompartments was assigned a segment-specific CL_{int} producing a maximum difference in the metabolic clearance from the PP to the PV region of 25.4-fold for DLZ and 18.7-fold for DPH (Table 5). The simulated blood, liver, and subcompartment concentration-time profiles for DLZ after intravenous or oral dosing with different enzyme zonation patterns are displayed in Fig. 13, and the resulting fold changes in PK parameters (T_{max} , AUC_{inf} and C_{max}) with hepatic zonation comparing to those obtained assuming even enzyme distribution are shown in Fig. 14. As can be seen, the blood PK are not sensitive to the change in zonation patterns regardless of the dosing route. In contrast, the total liver concentration-time profiles were markedly affected, for which the T_{max} , AUC_{inf} , and C_{max} are the highest when the metabolic activity is higher at the PV region and the lowest when the PP region exhibits the

higher metabolic clearance. In general, the changes in C_{max} after oral dosing are more noticeable than those after intravenous dosing when hepatic zonation is considered. The PK profile in each of the liver segments is affected differently by metabolic zonation, and such differences are also dependent on the dosing route. Among all of the subcompartments, the last one is the least affected. For the remaining subcompartments, lower CL_{int} at the PP region yields higher T_{max} , with the largest increase observed in the third compartment for the intravenous dose and the first compartment for the oral dose, whereas lower PV CL_{int} produces lower T_{max} , with the largest decrease occurring in the first subcompartment for both intravenous and oral doses as compared with the uniformly distributed CL_{int} . The C_{max} and AUC_{inf} show similar trends but more remarkable changes with differed zonation patterns than those observed for T_{max} . Such impacts of metabolic zonation were found to be similar for DPH Supplemental Figs. 5 and 6).

Discussion

The mPBPK models serve as a semimechanistic top-down approach between full PBPK models and empirical compartment models when only the blood/plasma PK data are available (the most common situation in human studies). These models have demonstrated broad applications in both preclinical and clinical research, owing to the flexibility of handling different mechanisms in PK, and offer reduced complexity compared with full models. Thus far, the WSM has been the most applied liver structural model in both mPBPK and full PBPK models despite it being the least physiologically relevant. In contrast, the application of the more physiologically plausible DM has been limited by its computational complexities. As shown previously, the SCM closely approximated the DM in PBPK as it yields comparable values and consistent model dependencies of CL_{int} and K_{ph} with those obtained by the DM (Li and Jusko, 2022, 2023). More importantly, it offers insights regarding the sequential intrahepatic concentration gradients and metabolic heterogeneity. Therefore, the current work demonstrates the implementation of the SCM in mPBPK models as an alternative to the DM and further expands the utility of mPBPK models for assessing

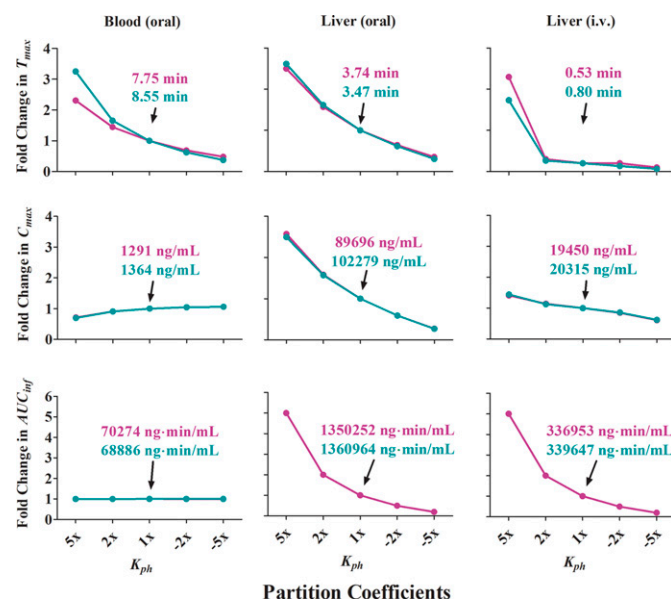


Fig. 10. Fold changes in the PK parameters calculated based on the PK profiles for DLZ simulated with varying values of K_{ph} displayed in Fig. 9. Different colors represent the results obtained by the mPBPK model extended with SCM-1 (purple) or SCM-5 (green). The nominal PK parameter values at $1 \times K_{ph}$ are listed with the same color coding as that for the indicated model.

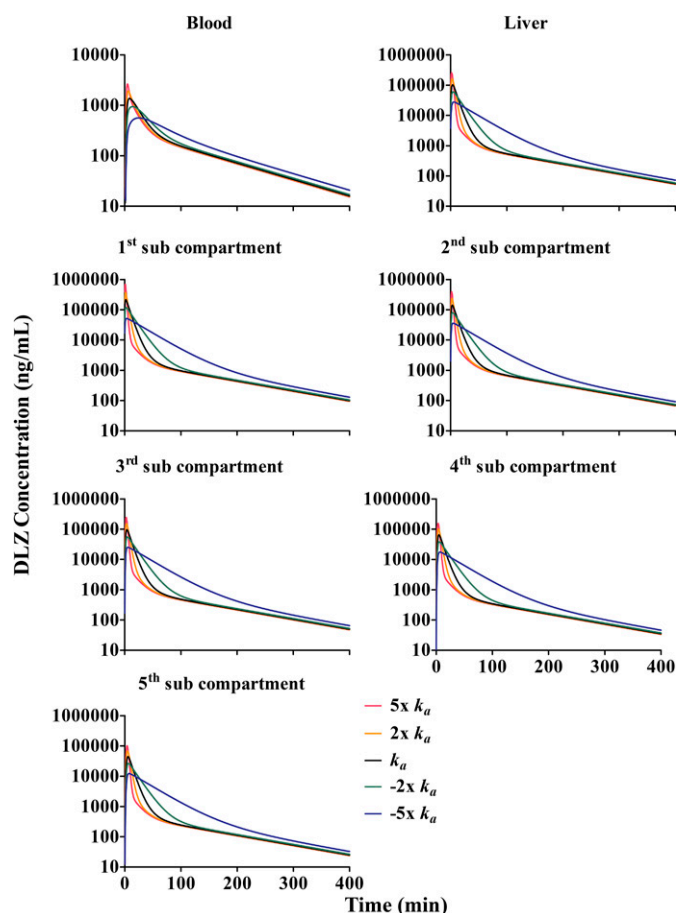


Fig. 11. Effects of changing k_a on predicting the blood, liver, and hepatic sub-compartment concentration-time profiles for DLZ after oral dosing by the mPBPK model extended with the SCM ($n = 5$). Color coded lines indicate model simulations with k_a (0.125 minutes⁻¹, black), -2 -fold k_a (green), $+2$ -fold k_a (orange), -5 -fold k_a (red), and $+5$ -fold k_a (blue).

hepatic distribution and clearance, oral absorption, and first-pass effects using the literature-reported blood and liver PK data for several substances mainly cleared by the liver after single intravenous and oral dosing.

The previous assessments of different hepatic models in PBPK (Li and Jusko, 2022, 2023) adopted a stepwise open-loop approach (Ebling et al., 1994; Cheung et al., 2018) where the blood PK was fitted first using empirical exponential functions to obtain the hepatic ER and model-specific CL_{int} , which were then fixed for fitting the liver PK to only estimate K_{ph} . Here, operating the SCM in mPBPK allows joint fitting of the blood and liver PK data and simultaneous estimation of the hepatic ER , CL_{int} , and K_{ph} and generates comparable results with those obtained by the open-loop approach (Fig. 3). Additionally, the blood and liver PK data for all 14 model compounds after intravenous dosing were more naturally captured, and the previous underpredictions for the liver PK of CyA and EB by the DM applying the open-loop approach were significantly improved (Fig. 2). Early time oscillations particularly seen in the mixed venous and arterial PK profiles after intravenous bolus administration are one of the distinct features of the whole-body PBPK models incorporated with the DM, with the lower D_N yielding more significant divergence from that obtained by the WSM (Oliver et al., 2001). Interestingly, the early phases of the SCM-fitted blood PK profiles for some of the compounds also exhibited similar oscillatory features and diverged from that predicted by the WSM (i.e., when $n = 1$ for the SCM), and such a divergence becomes more significant as n increases (Supplemental Fig. 7). This was not observed in the previous

assessments of the SCM, as the blood PK were fixed (Li and Jusko, 2023). Such findings are consistent with the expected quantitative relationships between the SCM and DM (i.e., the SCM with increasing n quantitatively correlates with the DM with decreasing D_N) (Li and Jusko, 2023).

The utility of the extended mPBPK models in assessing the oral absorption kinetics and first-pass effects was demonstrated by jointly fitting the blood and liver PK data for 12 model compounds after intravenous and oral dosing, with all of the PK data well captured and parameters precisely estimated. As was shown, the extended mPBPK models not only allow for characterization of nonlinear binding and metabolism in the liver but are also flexible in handling different sites of administration as well as the oral absorption kinetics either simply as a first-order process or applying the transit compartment models to account for absorption delays. Given that mPBPK models retain key physiologic components of the full PBPK models, it is readily feasible to incorporate more mechanistic absorption models as can be done with full models when relevant information is available (Huang et al., 2009; Cheng and Wong, 2020). The prehepatic availability (F_g) (i.e., the fraction of oral dose escaping prehepatic loss) was nicely informed and precisely estimated after the inclusion of oral data, with the estimated hepatic ER comparable to those obtained by fitting the intravenous data alone. The F_g is a hybrid parameter representing the product of the fraction of dose absorbed and the fraction escaping gastrointestinal (GI) extraction [metabolism and/or P-glycoprotein (P-gp)], the exact value of which can be further identified with available information pertaining to the PK processes happening before the orally dosed drug reaches the liver. Even though the selected compounds are mainly cleared by the liver, some of them showed noticeable prehepatic loss as reflected by low F_g values. This could be explained by poor absorption due to the solubility/dissolution issues if the GI extraction is negligible. This may not be true for CyA (Ducharme et al., 1995), QD (Darbar et al., 1997), and VEM (Darbar et al., 1998), as their intestinal metabolism was reported to be appreciable after oral doses. QD and VEM are known substrates of P-gp; nonetheless, the contribution of P-gp to the absorption and overall oral bioavailability of these compounds was considered limited given their high permeability and the easily saturable feature of intestinal P-gp, especially at the doses assessed herein (Varma and Panchagnula, 2005; Mori et al., 2012). It is expected that absorption

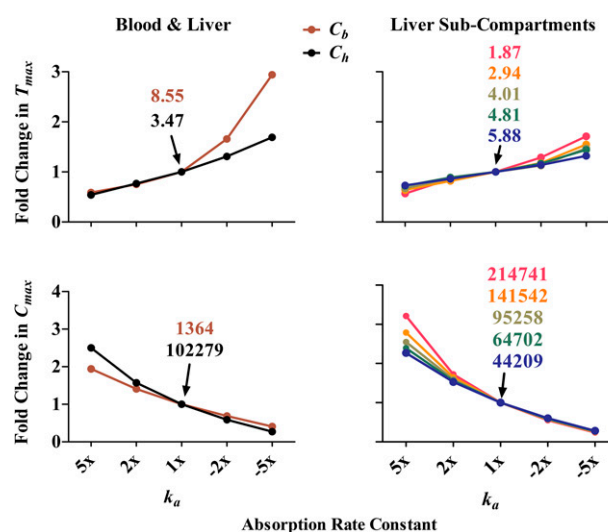


Fig. 12. Fold changes in the PK parameters calculated based on the PK profiles for DLZ simulated with varying k_a values shown in Fig. 11. Different compartments are color coded as indicated. The nominal parameter values at $1 \times k_a$ are listed with the same color coding as that used for the corresponding compartments.

TABLE 5

Values of hepatic intrinsic clearance (ml/min/kg) for DPH and DLZ with different patterns of metabolic enzyme zonation in a five-compartment liver model

Compound	ER ^a	Metabolism Pattern	$f_{ub}CL_{int1}^b$	$f_{ub}CL_{int2}^b$	$f_{ub}CL_{int3}^b$	$f_{ub}CL_{int4}^b$	$f_{ub}CL_{int5}^b$	$f_{ub}CL_{int}^b$
DLZ	0.83	Lower PP CL_{int}	3.58	7.37	15.62	35.26	90.97	152.8
		Lower PV CL_{int}	90.97	35.26	15.62	7.37	3.58	
		Even distribution			25.6 ^c			128
DPH	0.46	Lower PP CL_{int}	1.22	2.47	5.03	10.48	22.77	41.98
		Lower PV CL_{int}	22.77	10.48	5.03	2.47	1.22	
		Even distribution			7.92 ^c			39.6

^aThe ER for DPH and DLZ was from Tables 2 and 3.^bThe $f_{ub}CL_{int}$ ($i = 1, 2 \dots 5$) for uneven enzyme distribution was calculated from ER using eqs. 3, 4, and 20–23 and fixed in the model fitting of liver data; $f_{ub}CL_{int}$ is the sum of $f_{ub}CL_{int}$ of all liver subcompartments.^cThe $f_{ub}CL_{int}$ ($i = 1, 2 \dots 5$) for even enzyme distribution is calculated from $f_{ub}CL_{int}/5$.

and bioavailability would differ with different oral formulations, and this work demonstrated how such formulation-specific PK characteristics can be easily differentiated by the extended mPBPK models utilizing the oral PK data for QD in two different dosage forms (i.e., solution and powder in HPMC capsules).

In recent years, commercial software such as GastroPlus PBPK Simulator (Simulation Plus Inc., Lancaster, CA), PKSim (Open Systems Pharmacology Suite; Bayer Technology Services, Leverkusen, Germany), and Simcyp (Certara UK Ltd., Simcyp Division, Sheffield, UK) have made the implementation of PBPK modeling and simulation easier utilizing input parameter values obtained from various sources (Sager

et al., 2015; Lin et al., 2022). Specifically, the drug distribution parameters (e.g., K_{ph}) are often predicted by in silico approaches (Poulin and Theil, 2002; Berezhkovskiy, 2004; Rodgers and Rowland, 2006; Lukacova et al., 2008; Asaumi et al., 2019) based on the physicochemical properties of the drug substance and tissue compositions. However, abundant evidence has shown that such in silico-predicted K_{ph} values can differ appreciably from known in vivo values (Li et al., 2017; Song et al., 2020; Liu and Jusko, 2021; Jeong and Jusko, 2022; Li and Jusko, 2022). Here, we demonstrated that different K_{ph} values can result in markedly different predictions for the liver PK even though the intravenous blood PK are well matched (Figs. 9 and 10).

Both the extent and rate of absorption play critical roles in determining the bioavailability of orally administered drugs and are affected by many factors (e.g., the system physiologic parameters, the physicochemical properties and formulation characteristics of the drug, and concomitant consumption of food or other medications (Lennernäs, 1998; Martinez and Amidon, 2002; Cheng and Wong, 2020)). The present work further explored the impact of altering k_a on the oral PK characteristics. Not surprisingly, all oral PK profiles were sensitive to the changes in k_a (Figs. 11 and 12), with more remarkable increases in C_{max} for the first liver subcompartment as k_a increases. This is in line with predictions (Li and Jusko, 2023) that oral dosing with rapid absorption may exacerbate the differences in early model predictions and may be relevant when a drug shows hepatic toxicity.

It has been a common simplification in PBPK that metabolizing enzymes and transporters are uniformly distributed across the liver despite the recognition otherwise (Jungermann, 1986; Meyer-Wentrup et al., 1998; Oinonen and Lindros, 1998; Reichel et al., 1999; Tachikawa et al., 2018). Moreover, some hepatic enzymes are also regulated in a zonal pattern (Bühler et al., 1992; Lindros, 1997; Donner et al., 2004), which is considered to be associated with the zonal patterns of certain liver injuries (Cunningham and Porat-Shliom, 2021). For example, the regiospecific induction of CYP2E1 was reported to be the major reason for the centrilobular damage related to acetaminophen overdose (Anundi et al., 1993). The SCM offers needed flexibility in modeling liver zonation and regulation; however, related applications have been limited (Tirona and Pang, 1996; Abu-Zahra and Pang, 2000; Liu and Pang, 2006; Schenk et al., 2017; Li et al., 2019). Applying the extended mPBPK models with the SCM demonstrates incorporation of hepatic metabolic zonation and shows that differed zonation patterns can predict significantly different liver concentrations and intrahepatic concentration gradients even though the blood PK profiles are almost identical (Figs. 13 and 14).

The applications of the SCM are not only limited to the liver; for example, mechanistic absorption models such as the advanced compartmental absorption and transit (ACAT) model and the advanced dissolution, absorption, and metabolism (ADAM) model

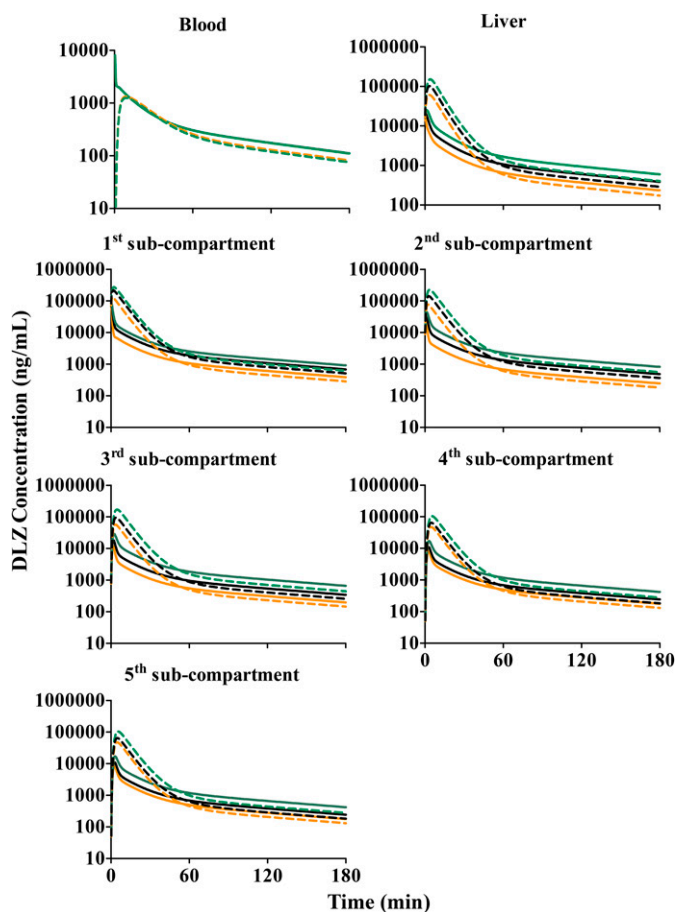


Fig. 13. Simulated concentration-time profiles for DLZ in blood, liver, and each of the liver subcompartments in rats after bolus administration of a 5-mg/kg i.v. dose (solid lines) or 20-mg/kg oral dose (dashed lines). Different hepatic zonation patterns are color coded as follows: lower PP CL_{int} (green), uniform CL_{int} (black), and lower PV CL_{int} (orange).

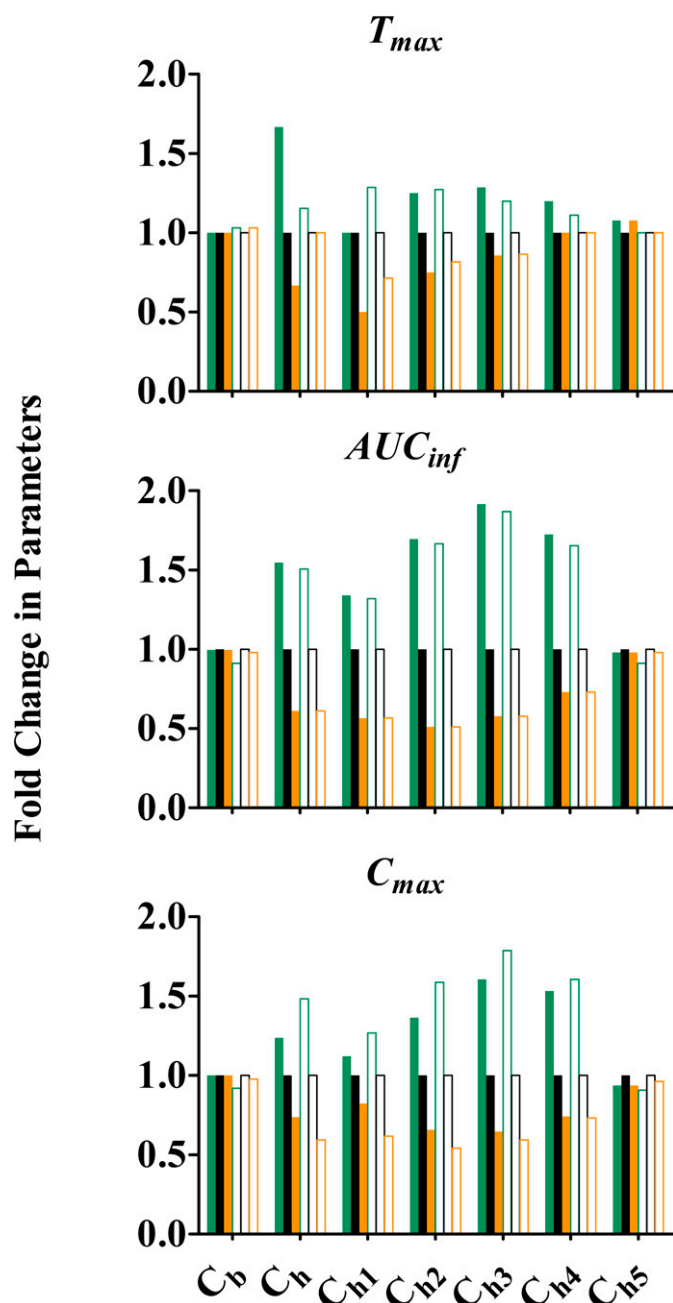


Fig. 14. Fold changes in T_{max} , AUC_{inf} , and C_{max} computed from the simulated PK profiles for DLZ in blood (C_b), liver (C_h), and each of the liver subcompartments (C_{h1} through C_{h5}) after intravenous (solid bars) and oral (open bars) administration with hepatic enzyme zonation [lower PP CL_{int} (green) or lower PV CL_{int} (orange)] compared with those obtained with uniform enzyme distribution (black).

treat the GI tract as a series of well mixed compartments with linear transfer kinetics and uniform concentrations (Huang et al., 2009). Recently, (semi)mechanistic kidney models have been developed to assess renal clearances where the kidney was divided into various nephron segments (Dave and Morris, 2015; Huang and Isoherranen, 2018). All of these models encompass the basic structure of the SCM.

One limitation of this work is that the intravenous and extravascular PK data were not available from the same literature sources except those for FTY720 and thus the parameter estimates are approximate due to potential data heterogeneity. Nevertheless, the results are deemed sufficient for demonstrating the expanded utilities of the extended mPBPK models.

In conclusion, the present work showcases the application of the SCM in mPBPK models and demonstrates that the extended mPBPK models are highly flexible in handling hepatic distribution and clearance, nonlinear tissue binding and metabolism, different oral absorption kinetics, and the relative contribution of prehepatic loss and hepatic extraction to the first-pass effects without the need for extensive PK data and tissue predictions such as required in full PBPK models. The extended mPBPK models with the SCM allow for the consideration of liver zonation and are particularly useful in assessing changes in the internal concentration gradients potentially masked by the similar or identical blood PK profiles due to alterations in drug- or system-specific factors. This is meaningful as intraorgan concentrations are considered the physiologically active moiety often driving metabolism, transport, pharmacodynamic (PD) effects, and toxicity. The mPBPK models extended with the SCM may be further appreciated with the technological advances in the study of liver zonation (Cunningham and Porat-Shliom, 2021; Paris and Henderson, 2022) and other aspects of intrahepatic functioning.

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Data Availability

The authors declare that all of the data supporting the findings of this study are contained within the paper.

Authorship Contributions

Participated in research design: Li, Jusko.

Performed data analysis: Li.

Wrote or contributed to the writing of the manuscript: Li, Jusko.

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